

UNIT V

NUCLEAR REACTIONS

Lesson Plan 5.1

Syllabus: Radioactive and stable isotopes - Mass defect and binding energy

Reference: Chemistry for Engineering students by Lawrence S Brown and Thomas A Holme (Fourth Edition) Chapter 13

INTRODUCTION

Nuclear chemistry deals with the study of composition of the nuclei of atoms, nuclear model, nuclear forces, radioactive materials and nuclear reactions. The energies involved in some of the nuclear reactions are million times greater than those in the chemical reactions. The nuclear energy holds out is great hope for the future as the present natural resources of energy are fast approaching exhaustion. The application of radioactive isotopes as ‘tracers’ has become an important tool in scientific and industrial developments.

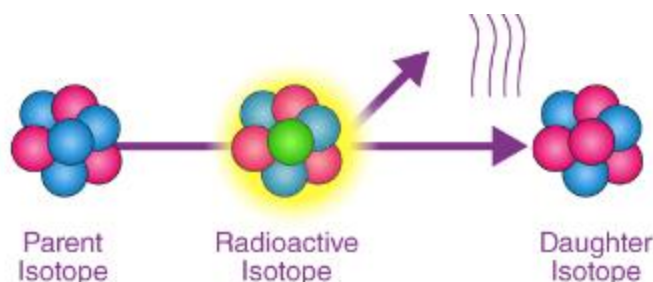
The fossil fuels are now conventional sources of power generation. In chemical fuels the chemical energy stored is released in the form of heat and these chemical reactions involved only the rearrangement of outer electrons. The atomic nucleus remains intact during a chemical reaction. Due to the exploitation of fossil fuels, they will not last for more than a few decades to supply the demand of electric power. Now attention has been focused on the exploitation of nuclear energy and other sources of power like solar energy, wind and tides for producing power.

During a nuclear reaction the number of protons and neutrons in a nucleus is altered. The product is a new nucleus of another atom with a different atomic number and mass number. So a nuclear reaction is one which proceeds with a change in the composition of the nucleus so as to produce an atom of a new element. The energy released when the nuclei of certain atoms undergo induced fission or fusion are very much greater than that obtained from chemical fuels. As the fossil fuels are fast approaching exhaustion, the release of nuclear energy during a nuclear reaction holds out great hope for the future. Table 1 shows the difference between nuclear and chemical reactions.

Differences between nuclear reactions and chemical reactions

S.No	Nuclear reactions	Chemical reactions
1	Nuclear reactions involve a change in the composition of the nucleus	Chemical reactions involve the rearrangement of outer electrons.
2	One element is converted into another, i.e. a new element is produced	No production of new element.
3	Reactions are accompanied by release or absorption of a large amount of energy.	Reactions are accompanied by release or absorption of relatively a small amount of energy
4	Rate of reaction is not influenced by external factors like concentration, temperature, pressure and catalyst.	Rate of reaction is influenced by external factors.

RADIOACTIVITY



Atomic theory in the nineteenth century presumed that nuclei had fixed compositions. But in 1896, the French scientist Henri Becquerel found that a uranium compound placed near a photographic plate made an image on the plate, even if the compound was wrapped in black cloth. He reasoned that the uranium compound was emitting some kind of radiation that passed through the cloth to expose the photographic plate. Further investigations showed that the radiation was a combination of particles and electromagnetic rays, with its ultimate source being the atomic nucleus. These emanations were ultimately called, collectively, **radioactivity**.

Following the somewhat serendipitous discovery of radioactivity by Becquerel, many prominent scientists began to investigate this new, intriguing phenomenon. Among them were Marie Curie (the first woman to win a Nobel Prize, and the only person to win two Nobel Prizes in different sciences—chemistry and physics), who was the first to coin

the term “radioactivity,” and Ernest Rutherford (of gold foil experiment fame), who investigated and named three of the most common types of radiation. During the beginning of the twentieth century, many radioactive substances were discovered, the properties of radiation were investigated and quantified, and a solid understanding of radiation and nuclear decay was developed.

The spontaneous change of an unstable nuclide into another is radioactive decay. The unstable nuclide is called the parent nuclide; the nuclide that results from the decay is known as the daughter nuclide. The daughter nuclide may be stable, or it may decay itself. The radiation produced during radioactive decay is such that the daughter nuclide lies closer to the band of stability than the parent nuclide, so the location of a nuclide relative to the band of stability can serve as a guide to the kind of decay it will undergo (Figure 1). The alpha particle removes two protons (green) and two neutrons (gray) from the uranium-238 nucleus.

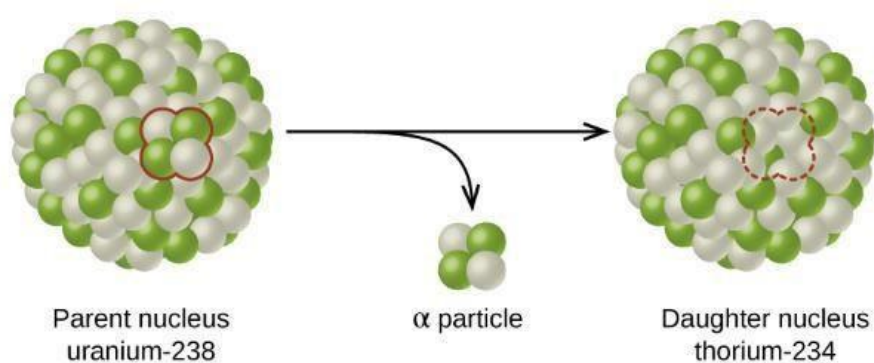
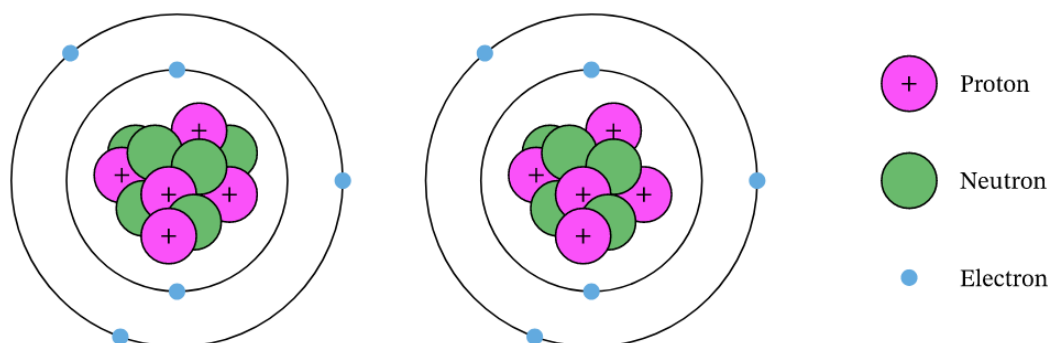


Figure 1: A nucleus of uranium-238 (the parent nuclide) undergoes α decay to form thorium-234 (the daughter nuclide).

ISOTOPES:

The term “isotope” mainly refers to the variation in the atomic mass or weight of an element. It is also defined as variants of a particular element where these variants will have the same number of protons but differ in the number of neutrons in the atom.



Due to the unequal numbers of neutrons, the isotopes of elements usually have a different mass. Generally, elements which have odd atomic numbers will have one or two stable isotopes

whereas elements with even atomic numbers will mostly have 3 or more stable isotopes. However, there are also exceptions like carbon, helium, and beryllium.

An isotope is usually denoted or identified by the name of the particular element at the beginning which is followed by a hyphen and the mass number.

Physical and Chemical Properties of Isotopes

The chemical properties of isotopes of a given element are almost the same or identical. Different isotopes exhibit nearly identical chemical behaviors. However, when it comes to physical properties of isotopes like the mass, melting or boiling point, density, and freezing point they all are different.

The physical properties of any isotope mostly depend on the mass of each isotope. Knowing the differences allows us to distinguish between one isotope from another.

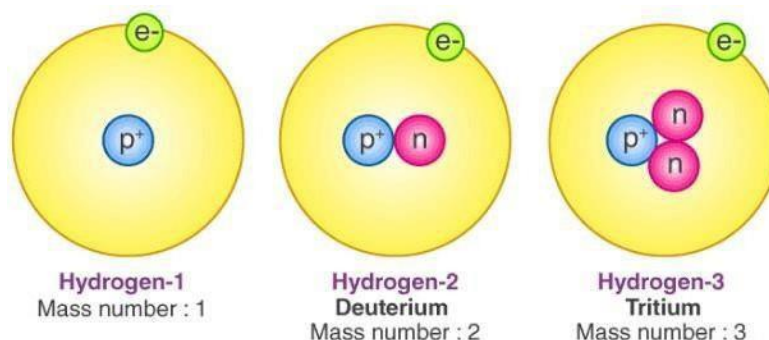
Isotopes are said to be either stable or radioactive. Therefore, isotopes that are radioactive are often called radioisotopes or radionuclides. Isotopes that do not decay radioactively are known as stable isotopes or stable nuclides.

As per the findings, there are about 339 naturally occurring nuclides or isotopes on the planet earth. In this, 286 are said to be primordial nuclides which are believed to have existed since the formation of the Solar System.

Examples

The common examples are the isotopes of hydrogen and carbon. If we talk about the element Hydrogen, it has three stable isotopes namely protium, deuterium, and tritium.

Hydrogen is the first element in the periodic table and has the atomic number one. Those elements which have the same atomic number but a different mass number are called isotopes. There are three isotopes of hydrogen namely, protium ${}^1_1\text{H}$, deuterium ${}^2_1\text{H}$ or D and lastly tritium ${}^3_1\text{H}$ or T. The isotopes are different because of the different number of neutrons present in them. In protium, there is no presence of neutrons, whereas in deuterium we have one neutron and in tritium, we have two neutrons. The most prominent form of hydrogen is protium, 0.0156% of hydrogen is present on the earth's surface as deuterium. In tritium, the concentration is one atom per 10¹⁸ atoms of protium.



Isotopes of Hydrogen

Out of these three isotopes of hydrogen, only tritium is radioactive in nature which emits low-energy beta particles. As the electronic configuration of isotopes is the same, they all have similar chemical properties. But they have a difference in their rates of reaction, this happens because of the different bond disassociation enthalpies. They have different physical properties because of the large variations in mass.

The occurrence of hydrogen on earth is difficult due to its light nature. In combined it forms 15.4% of hydrogen that occurs in the earth's crust and oceans. Hydrogen also occurs in plant and animal tissues, hydrocarbons, proteins, hydrides, and many other compounds. In the universe, hydrogen is the most abundant element (70% of the total mass) and it is also the principal element of the solar atmosphere. Even the huge planets such as Jupiter and Saturn mainly consist of hydrogen.

(a) Protium (^1H)

It is one of the common isotopes of hydrogen. It is plenty in nature with an abundance of 99.98%. One of the reasons for this is that the nucleus of this isotope consists of a single proton and this proton at no time, it has been reported to be decayed. Mass of protium is 1.007825 amu. Hydrogen generally combines with other atoms in compounds and are usually found in H_2 (diatomic hydrogen gas).

(b) Deuterium (^2H)

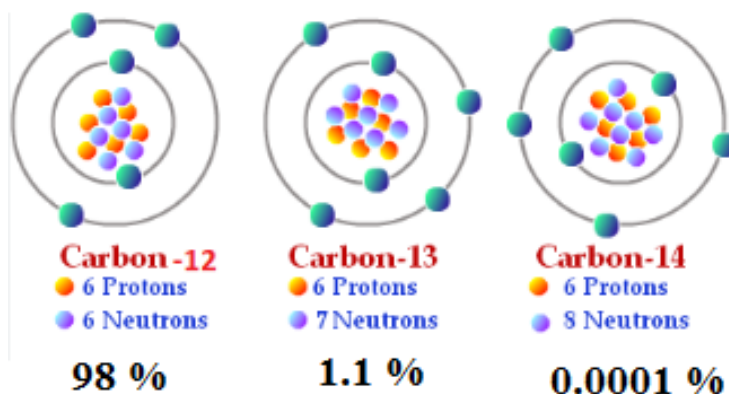
It comprises 1 proton and 1 neutron in its nucleus. The nucleus of hydrogen 2 is termed as deuteron. It is not radioactive. Its compounds are used in chemical analysis and solvents for hydrogen 1. Heavy water is enriched with molecules consisting of deuterium instead of protium. It used as a coolant and a neutron moderator. Hydrogen 2 is also used as a fuel in nuclear fusion (commercial). It occurs naturally as deuterium gas.

(c) Tritium (^3H)

It comprises 2 neutrons and 1 proton in its nucleus. Small traces of hydrogen 3 or tritium occurs in nature due to the synergy of cosmic rays with atmospheric gases. They are also released in a small amount at the time of nuclear weapons tests. It is radioactive, it decays into helium 3 through beta decay. Hydrogen 3 has an atomic mass of 3.0160492 u.

Isotopes of Carbon

Carbon isotopes come in three forms. By far the most common isotope of carbon is carbon-12 (^{12}C), which contains six neutrons in addition to its six protons. The next heaviest carbon isotope, carbon-13 (^{13}C), has seven neutrons. Both ^{12}C and ^{13}C are called stable isotopes since they do not decay into other forms or elements over time. The rare carbon-14 (^{14}C) isotope contains eight neutrons in its nucleus. Unlike ^{12}C and ^{13}C , this isotope is unstable, or radioactive. Over time, a ^{14}C atom will decay into a stable product.



The vast majority of all carbon found on Earth is ^{12}C . Almost 99% of all carbon on Earth is of this form. While only approximately 1% of all carbon on Earth is of the ^{13}C isotopic form, ^{14}C is still much rarer. Only one out of every trillion carbon atoms is ^{14}C .

Apart from these some other common isotope examples include – Tin has 22 isotopes, Zinc has 21 known isotopes, Neon is a mix of 3 isotopes, natural xenon consists of a mixture of 9 stable isotopes, and Nickel has 14 known isotopes.

ENERGY RELEASED IN NUCLEAR REACTIONS

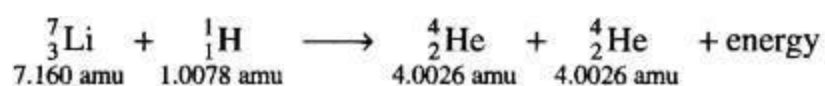
The energy changes associated with nuclear reactions are million times more than that of chemical reactions. These are generally determined by using Einstein's mass-energy relation.

$$E = mc^2$$

where E is the energy equivalent of mass m and c is the velocity of light. For a change of atomic mass unit (amu), the corresponding energy change:

$$\begin{aligned}
 E &= 1.6605 \times 10^{-27} \text{ kg} \times (2.9979 \times 10^8 \text{ ms}^{-1})^2 \\
 &= 1.4924 \times 10^{-10} \text{ kg m}^2 \text{ s}^{-2} = 1.4924 \times 10^{-10} \text{ J} \\
 &= 1.4924 \times 10^{-10} \text{ J} \times \left(\frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} \right) \\
 &= 931 \times 10^6 \text{ eV} = 931 \text{ MeV}
 \end{aligned}$$

That is a mass of 1 amu is equivalent to 931 Me V of energy. Very often in nuclear reactions the mass of the products is less than that of the reactants, The mass difference is converted into energy. For example:



In the above equation, the atomic mass difference between the reactants and products is 0.1626 amu. The energy released during this nuclear reaction is calculated by the following simple relation:

$$\begin{aligned}
 \frac{931 \text{ MeV}}{1 \text{ amu}} &= \frac{x \text{ MeV}}{0.1626 \text{ amu}} \\
 x &= 931 \times 0.1626 \text{ MeV} \\
 &= 151.4 \text{ MeV}
 \end{aligned}$$

Thus in general the nuclear energy can be determined by multiplying the loss in mass by 931 MeV.

Mass Defect (Δm)

Mass defect may be defined as the difference between the mass of an atom and sum of the free masses of its protons and neutrons (collectively known as nucleons). For example, the mass defect of helium atom having two protons (mass of one proton = 1.00815 amu) and two neutrons (mass of one neutron = 1.00899 amu) with the measured mass of 4.0026 is calculated as follows:

$$\text{Total expected mass of helium} = 2 \times 1.00815 + 2 \times 1.00899$$

$$= 2.0163 + 2.01798$$

$$= 4.03428 \text{ amu.}$$

The mass defect always represents the amount of disappearance of matter during the formation of a nucleus by packing together the protons and neutrons.

Binding Energy

The binding energy is a measure of the force that holds the nucleons (protons and neutrons) together. In a nucleus the positively-charged protons and uncharged neutrons are bound to one another by very powerful forces. The energy that binds the nucleons together in a nucleus is called the binding energy. The binding energy is usually expressed in terms of million electron volts (MeV). In other words, energy equal to binding energy will be needed to break up the nucleus into its constituent nucleons. Consequently, the greater is the binding energy, the more stable is the nucleus. When a nucleus is formed from individual protons and neutrons, there occurs a loss of mass (mass defect). According to Einstein's theory it is this mass defect which is converted into binding energy. Hence binding energy is the energy equivalent to the mass defect. Mathematically, the binding energy (E_B) and the mass defect (Δm) are related as follows:

$$E_B = 931 \times \Delta m$$

Binding energy per nucleon can be calculated by dividing the total binding energy by the sum of the number of protons and neutrons present in a nucleus, i.e.

$$\text{Binding energy per nucleon} = \frac{\text{Binding energy of nucleus}}{\text{Number of protons} + \text{Number of neutrons}}$$

Binding Energy and Nuclear Stability:

By plotting the binding energy per nucleon against the mass number, a graph is obtained. This shows the relative stability of various nuclei.

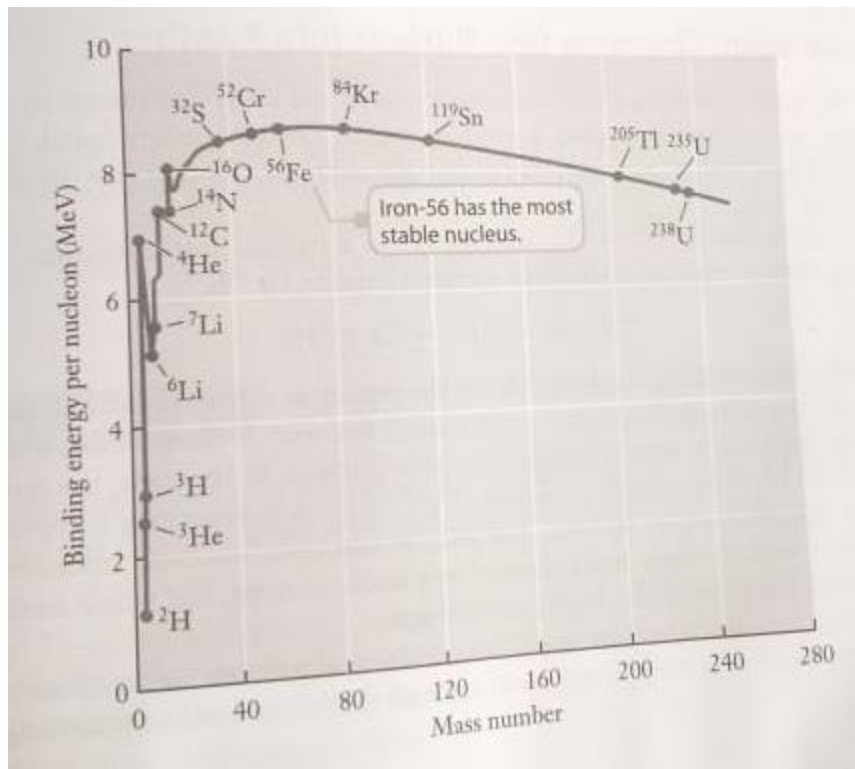


Fig. Relative stability of various nuclei.

The greater the binding energy per nucleon, the more stable is the nucleus. Thus the nuclei of mass number 60 having maximum energy per nucleon are more stable, e.g. ^{56}Fe . The nuclei that are heavier and lighter than this have lower binding energies per nucleon and are less stable. Thus ^{235}U undergoes fission to give lighter and more stable isotopes such as ^{139}Ba and ^{94}Kr with the release of energy. Similarly, two or more lighter nuclei (^2H , ^1H) with lower binding energy per nucleon combine together to give a heavier and more stable nucleus with the release of energy.

SOLVED PROBLEMS

1. Calculate the binding energy in MeV of ^4_2He which has a mass of 4.00390 amu. The masses of a proton, a neutron and an electron are 1.007825, 1.008668 and 0.0005852 amu respectively.

Solution: ${}^4_2\text{He}$ has two protons, two electrons and two neutrons.

$$\begin{aligned}\text{Expected mass of helium} &= \text{mass of protons} + \text{mass of neutrons} + \text{mass of electrons} \\ &= 2 \times 1.007825 + 2 \times 1.008668 + 2 \times 0.0005852 \\ &= 2.01565 + 2.017336 + 0.0011704 \\ &= 4.0341564 \text{ amu}\end{aligned}$$

$$\text{Measured mass of helium} = 4.00390 \text{ amu}$$

$$\begin{aligned}\text{Mass defect} &= 4.0341564 - 4.00390 \\ &= 0.0302564 \text{ amu}\end{aligned}$$

$$\begin{aligned}\text{Binding energy released per nucleus} &= \text{mass defect} \times 931 \text{ MeV} \\ &= 0.0302564 \times 931 \\ &= 28.17 \text{ MeV}\end{aligned}$$

2. Calculate the mass defect and binding energy per nucleon for ${}^{59}_{27}\text{Co}$ given that the atomic masses of cobalt, hydrogen atom and neutron are 58.95182 amu, 1.008142 amu and 1.008982 amu respectively.

Solution:

$$\text{Mass number of cobalt} = 59$$

$$\text{Number of protons in cobalt or atomic number of cobalt} = 27$$

$$\begin{aligned}\text{Number of neutrons in cobalt} &= \text{Mass number of cobalt} - \text{Atomic number of cobalt} \\ &= 59 - 27 = 32\end{aligned}$$

$$\text{Mass defect } (\Delta m) = \text{Expected mass of cobalt} - \text{measured mass}$$

$$\begin{aligned}\text{Expected mass of cobalt} &= \text{mass of protons} + \text{mass of neutrons} \\ &= 27 \times 1.008142 + 32 \times 1.008982 \\ &= 27.219834 + 32.287424 \\ &= 59.507258\end{aligned}$$

$$\text{Measured mass of cobalt} = 58.95182$$

$$\begin{aligned}\text{Mass defect} &= 59.507258 - 58.95182 \\ &= 0.555438 \text{ amu}\end{aligned}$$

$$\text{Binding energy of the nucleus} = 931 \times 0.555438$$

$$\begin{aligned}\text{Binding energy per nucleon} &= \frac{931 \times \text{mass defect}}{\text{mass no. of cobalt}} \\ &= \frac{931 \times 0.555438}{59} \\ &= 8.76 \text{ MeV}\end{aligned}$$

3. Problem: Calculate the energy released by the nucleus Uranium-235 if it splits into barium-141 and a krypton-92 nucleus.

Solution The masses of the various particles involved are shown in the following table:

Particle	Mass (u)
^{235}U	235.0439231
^{141}Ba	140.9144064
^{92}Kr	91.9261528
^1_0n	1.0086649

We can sum the appropriate particle masses to find the masses of the reactants and the products:

$$\begin{aligned}\text{Mass of reactants} &= 235.0439231 \text{ u} + 1.0086649 \text{ u} \\ &= 236.0525880 \text{ u}\end{aligned}$$

$$\begin{aligned}\text{Mass of products} &= 140.9144064 \text{ u} + 91.9261528 \text{ u} + 3(1.0086649 \text{ u}) \\ &= 235.8665539 \text{ u}\end{aligned}$$

Now we subtract to find the mass defect.

$$\begin{aligned}\Delta m &= 235.8665539 \text{ u} - 236.0525880 \text{ u} = -0.1860341 \text{ u} \\ E &= (\Delta m)c^2 \\ &= -0.1860341 \text{ u} \left(\frac{1.66053886 \times 10^{-27} \text{ kg}}{\text{u}} \right) \left(\frac{2.99792458 \times 10^8 \text{ m}}{\text{s}} \right)^2 \\ &= -2.776406 \times 10^{-11} \text{ kg m}^2/\text{s}^2 \\ &= -2.776406 \times 10^{-11} \text{ J}\end{aligned}$$

Discussion questions

1. Justify why Helium-4 is considered a highly stable nucleus compared to other light elements.
2. Identify the error in the following nuclear reaction:

$$1\text{H}_1 + 1\text{H}_1 \rightarrow 3\text{He}_2$$
3. Identify the principle behind the lost mass in a nuclear reaction (mass defect) and justify how it is transformed into energy.
4. Find the factors that determine why some elements have stable isotopes while others are radioactive.
5. Identify the role of the neutron-to-proton ratio in nuclear stability.